Experiment 1: Node.js HTTP Server

Code:

```
const http = require('http');

const server = http.createServer((req, res) => {
    const url = req.url;

    if (url === '/') {
        res.writeHead(200, { 'Content-Type': 'text/html' });
        res.write(`
            <!DOCTYPE html>
            <html>
            <head>
                <title>Home Page</title>
                <style>
                    body { font-family: Arial; text-align: center; padding: 50px; }
                    h1 { color: blue; }
                </style>
            </head>
            <body>
                <h1>Welcome to Home Page</h1>
                <p>This is the home page of my Node.js server</p>
                <a href="/about">Go to About</a> |
                <a href="/contact">Go to Contact</a>
            </body>
            </html>
        `);
        res.end();
    }
    else if (url === '/about') {
        res.writeHead(200, { 'Content-Type': 'text/html' });
        res.write(`
            <!DOCTYPE html>
            <html>
            <head>
                <title>About Page</title>
                <style>
                    body { font-family: Arial; text-align: center; padding: 50px; }
                    h1 { color: green; }
                </style>
            </head>
            <body>
```

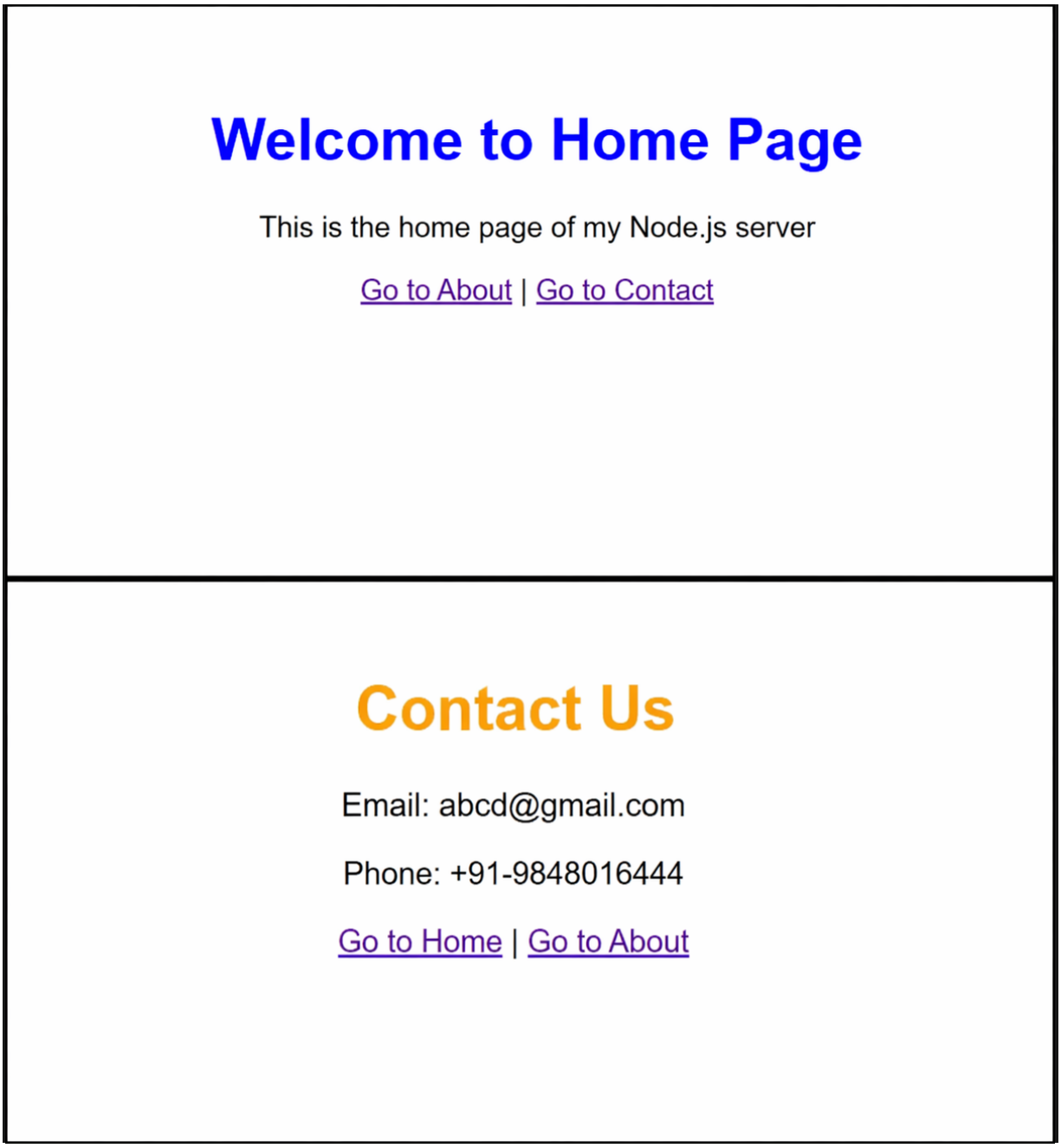
```
            <h1>About Us</h1>
            <p>I am ramu learning Node.js!</p>
            <p>Roll Number: 160124737001</p>
            <a href="/">Go to Home</a> |
            <a href="/contact">Go to Contact</a>
        </body>
        </html>
    `);

    res.end();
  }
  else if (url === '/contact') {
    res.writeHead(200, { 'Content-Type': 'text/html' });
    res.write(`
        <!DOCTYPE html>
        <html>
        <head>
            <title>Contact Page</title>
            <style>
                body { font-family: Arial; text-align: center; padding: 50px; }
                h1 { color: orange; }
            </style>
        </head>
        <body>
            <h1>Contact Us</h1>
            <p>Email: abcd@example.com </p>
            <p>Phone: +91-9848016444</p>
            <a href="/">Go to Home</a> |
            <a href="/about">Go to About</a>
        </body>
        </html>
    `);
    res.end();
  }
  else {
    res.writeHead(404, { 'Content-Type': 'text/html' });
    res.write('<h1>404 - Page Not Found</h1>');
    res.end();
  }
});

server.listen(3000, () => {
  console.log('Server running at http://localhost:3000/');
```

```
    console.log('Student:  ramu  (160124737001)');
});
```

Output:



Experiment 2: File System Operations

Code:

```
const fs = require('fs');

console.log('=== Experiment 2: File System Operations ===');
console.log('Student: Addagudi Sathvik (160124737098)\n');

const studentDetails = `Student Details
================
Name: sham
Roll Number: 160124737001
Course: B.Tech Information Technology
Semester: 4th
Date: ${new Date().toLocaleDateString()}
`;

fs.writeFile('data.txt', studentDetails, (err) => {
    if (err) {
        console.log('Error writing file:', err);
        return;
    }
    console.log('File "data.txt" created successfully!');

    fs.readFile('data.txt', 'utf8', (err, data) => {
        if (err) {
            console.log('Error reading file:', err);
            return;
        }
        console.log('\n--- File Content ---');
        console.log(data);

        const newData = `\nAdditional Info:
================
Skills: JavaScript, Node.js, Express.js
Project: Completed Node.js experiments
Status: In Progress\n`;

        fs.appendFile('data.txt', newData, (err) => {
            if (err) {
                console.log('Error appending data:', err);
                return;
            }
```

```
            console.log('✓ New data appended successfully!\n');

            fs.readFile('data.txt', 'utf8', (err, updatedData) => {
                if (err) {
                    console.log('Error reading file:', err);
                    return;
                }
                console.log('--- Updated File Content ---');
                console.log(updatedData);
            });
        });
    });
});
```

Output:

```
=== Experiment 2: File System Operations ===
Student: sham(160124737001)

File "data.txt" created successfully!

--- File Content ---
Student Details
================
Name:  Sham
Roll Number: 160124737001
Course: B.Tech Information Technology
Semester: 4th
Date: 4/20/2026

✓ New data appended successfully!

--- Updated File Content ---
Student Details
================
Name:  Sham
Roll Number: 160124737001
Course: B.Tech Information Technology
Semester: 4th
Date: 4/20/2026

Additional Info:
================
```

Skills: JavaScript, Node.js, Express.js
Project: Completed Node.js experiments
Status: In Progress

data.txt

```
Student Details
================
Name:  Sham
Roll Number: 160124737001
Course: B.Tech Information Technology
Semester: 4th
Date: 4/20/2026

Additional Info:
================
Skills: JavaScript, Node.js, Express.js
Project: Completed Node.js experiments
Status: In Progress
```

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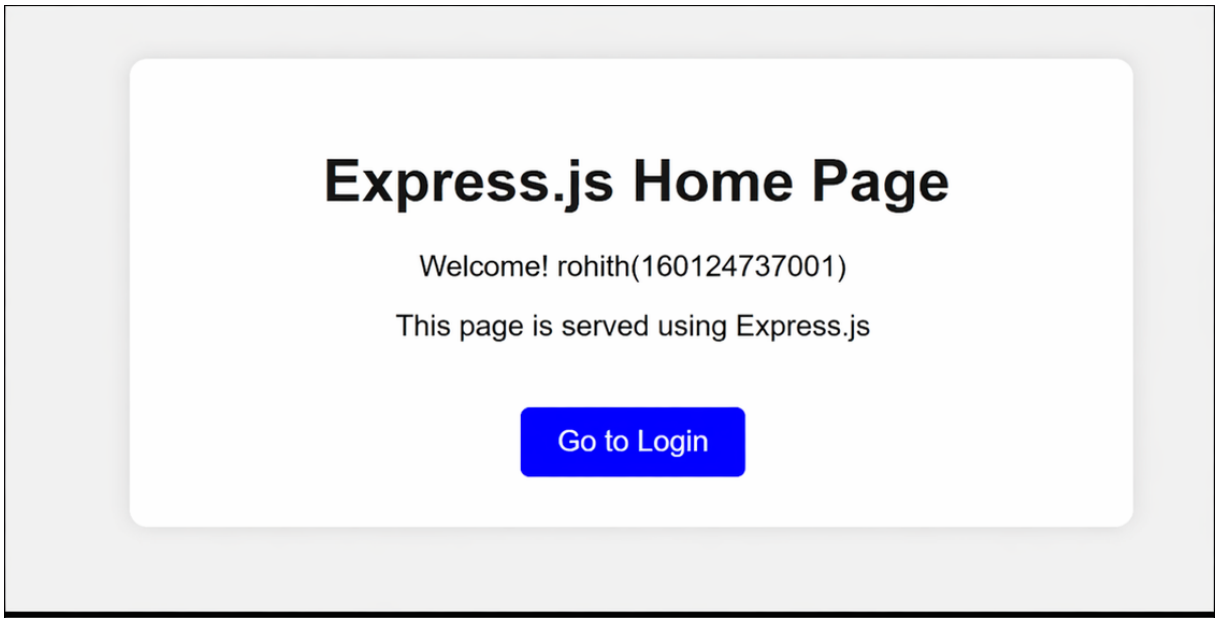
Experiment 3: Express.js Routing & Middleware

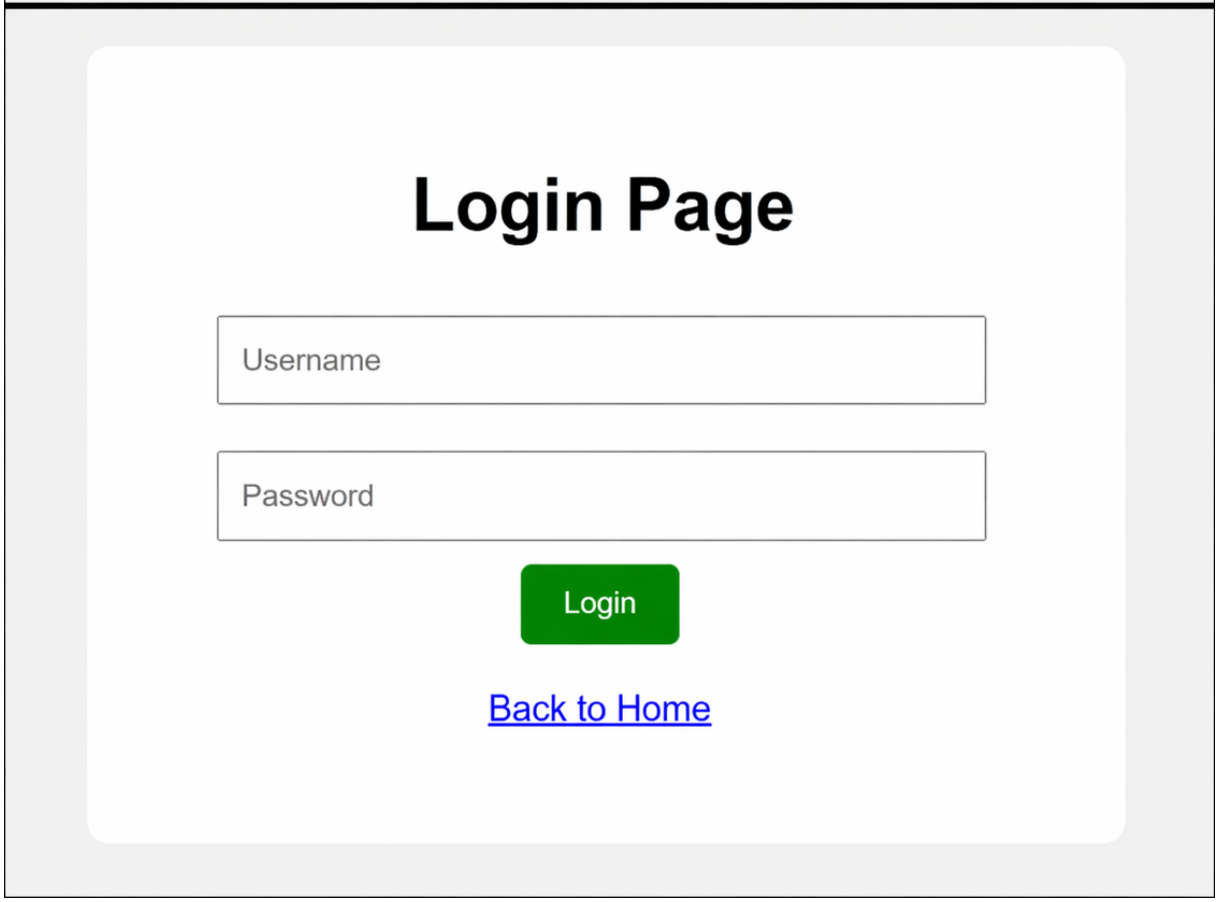
Code:

```
const express = require('express');
const path = require('path');
const app = express();
const PORT = 3000;

app.use((req, res, next) => {
    next();
});

app.get('/home', (req, res) => {
    res.send(`
        <!DOCTYPE html>
        <html>
        <head>
            <title>Home - Express App</title>
            <style>
                body { font-family: Arial; text-align: center; padding: 50px;
background: #f0f0f0; }
                .card { background: white; padding: 30px; border-radius: 10px;
max-width: 500px; margin: auto; box-shadow: 0 0 10px rgba(0,0,0,0.1); }
                h1 { color: #333; }
                .btn { background: blue; color: white; padding: 10px 20px; text-
decoration: none; border-radius: 5px; display: inline-block; margin-top:
20px; }
            </style>
        </head>
        <body>
            <div class="card">
                <h1>Express.js Home Page</h1>
                <p>Welcome! (160124737001)</p>
                <p>This page is served using Express.js</p>
                <a href="/login" class="btn">Go to Login</a>
            </div>
        </body>
        </html>
    `);
});


app.get('/login', (req, res) => {
    res.send(`
```

```
        <!DOCTYPE html>
        <html>
        <head>
            <title>Login Page</title>
            <style>
                body { font-family: Arial; text-align: center; padding: 50px;
background: #f0f0f0; }
                .card { background: white; padding: 30px; border-radius: 10px;
max-width: 400px; margin: auto; }
                input { padding: 10px; margin: 10px; width: 80%; }
                button { padding: 10px 20px; background: green; color: white;
border: none; border-radius: 5px; cursor: pointer; }
            </style>
        </head>
        <body>
            <div class="card">
                <h1>Login Page</h1>
                <form action="/login" method="POST">
                    <input type="text" placeholder="Username" name="username"
required><br>
                    <input type="password" placeholder="Password"
name="password" required><br>
                    <button type="submit">Login</button>
                </form>
                <p style="margin-top:20px"><a href="/home">Back to
Home</a></p>
            </div>
        </body>
        </html>
    `);
});

app.use(express.urlencoded({ extended: true }));
app.post('/login', (req, res) => {
    const { username, password } = req.body;
    console.log(`Login attempt: ${username}`);
    res.send(`
        <h1>Welcome ${username}!</h1>
        <p>Login successful (demo mode)</p>
        <a href="/home">Go to Home</a>
    `);
});
```

```
app.listen(PORT, () => {
    console.log(`- Home: http://localhost:${PORT}/home`);
    console.log(`- Login: http://localhost:${PORT}/login`);
});
```





Output:

Experiment 4: REST API using Express.js

Code:

```
const express = require('express');
const app = express();
const PORT = 3000;

app.use(express.json());

let students = [
    { id: 1, name: " Rohith", rollNumber: "160124737001", department:
"CSE", marks: 88 },
    { id: 2, name: "Rahul Sharma", rollNumber: "160124737101", department:
"CSE", marks: 92 },
    { id: 3, name: "Priya Patel", rollNumber: "160124737102", department:
"IT", marks: 85 }
];

let nextId = 4;

app.use((req, res, next) => {
    console.log(`[${new Date().toLocaleTimeString()}] ${req.method} ${req.url}`);
    next();
});

app.get('/students', (req, res) => {
    console.log(' Fetching all students');
    res.json({
        success: true,
        count: students.length,
        students: students
    });
});

app.get('/students/:id', (req, res) => {
    const id = parseInt(req.params.id);
    const student = students.find(s => s.id === id);

    if (!student) {
        return res.status(404).json({ success: false, message: 'Student not
found' });
    }
```

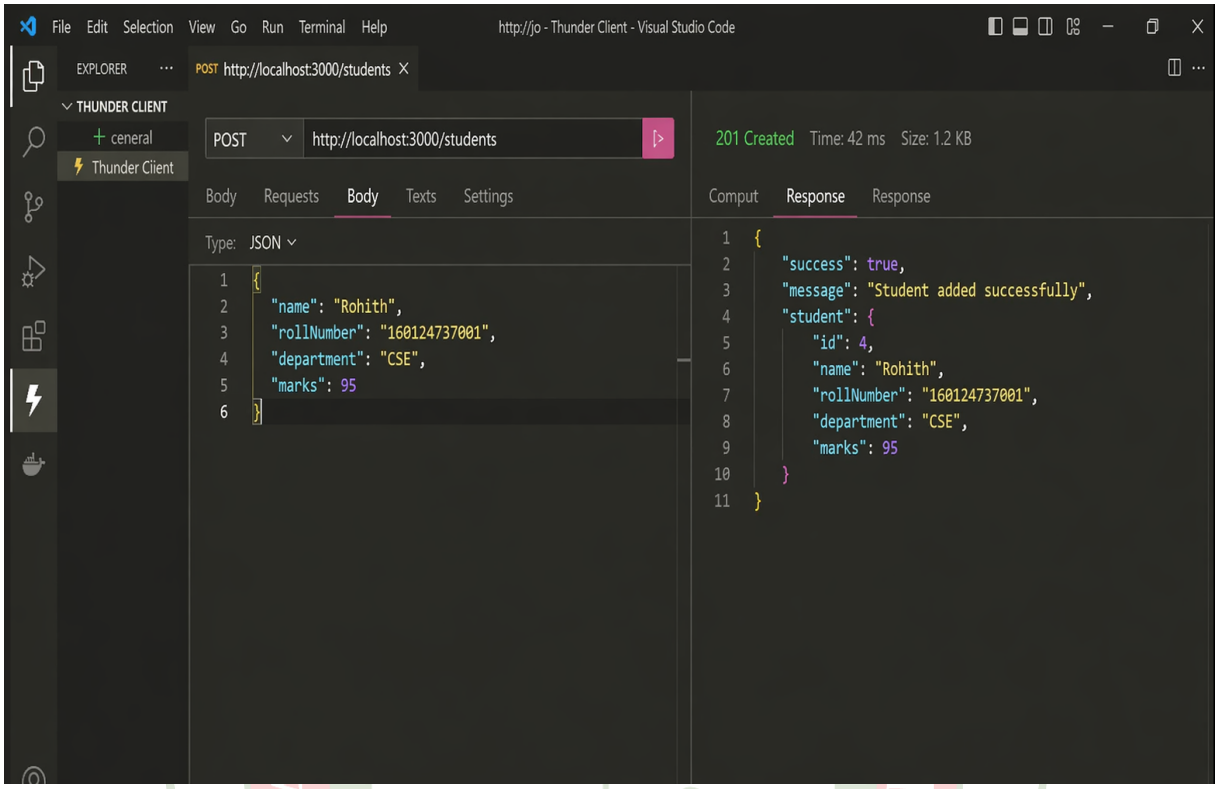
```
    res.json({ success: true, student });
});

app.post('/students', (req, res) => {
    const { name, rollNumber, department, marks } = req.body;

    if (!name || !rollNumber || !department || marks === undefined) {
        return res.status(400).json({
            success: false,
            message: 'Please provide name, rollNumber, department, and marks'
        });
    }

    const newStudent = {
        id: nextId++,
        name,
        rollNumber,
        department,
        marks
    };

    students.push(newStudent);
    console.log(` Added new student: ${name}`);

    res.status(201).json({
        success: true,
        message: 'Student added successfully',
        student: newStudent
    });
});

app.put('/students/:id', (req, res) => {
    const id = parseInt(req.params.id);
    const { name, rollNumber, department, marks } = req.body;

    const studentIndex = students.findIndex(s => s.id === id);

    if (studentIndex === -1) {
        return res.status(404).json({ success: false, message: 'Student not
found' });
    }
```

```
    students[studentIndex] = {
        ...students[studentIndex],
        name: name || students[studentIndex].name,
        rollNumber: rollNumber || students[studentIndex].rollNumber,
        department: department || students[studentIndex].department,
        marks: marks !== undefined ? marks : students[studentIndex].marks
    };

    console.log(` Updated student with id: ${id}`);

    res.json({
        success: true,
        message: 'Student updated successfully',
        student: students[studentIndex]
    });
});

app.delete('/students/:id', (req, res) => {
    const id = parseInt(req.params.id);
    const studentIndex = students.findIndex(s => s.id === id);

    if (studentIndex === -1) {
        return res.status(404).json({ success: false, message: 'Student not
found' });
    }

    const deletedStudent = students[studentIndex];
    students = students.filter(s => s.id !== id);

    console.log(` Deleted student: ${deletedStudent.name}`);

    res.json({
        success: true,
        message: 'Student deleted successfully',
        deletedStudent: deletedStudent
    });
});

app.listen(PORT, () => {
    console.log(`Server running at http://localhost:${PORT}`);
```

Laboratory Record
of Enterprise Application
Development Lab

Roll No.: 160124737095
Exp No.:
Sheet No.:
Date:

Output:



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